

Siddharth Khandelwal

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Education

Indian Institute of Technology Bombay, B.Tech in Electrical Engineering

2019 – 2023

- CGPA: 9.23/10
- Minor in Artificial Intelligence & Data Science
- Minor in Computer Science & Engineering

Publications

1. **S. Khandelwal**, S. Kamath, and A. Jain. “**Odo: Depth-Guided Diffusion for Identity-Preserving Body Reshaping.**” *ORAL Paper Presentation at Winter Conference on Applications of Computer Vision (WACV)*, 2026.
2. S. Banerjee, K. Sarda, **S. Khandelwal**, et al. “**IoT-based Sensing System for Thrips Pest and Disease Management in Onion Crop.**” *Best Paper Award at International IEEE Applied Sensing Conference*, 2024.
3. Y. Gupta, **S. Khandelwal**, et al. “**Lunar Exploration through Chipsats.**” *71st International Astronautical Congress (IAC), Cyberspace Edition*, 2020.

Work Experience

Research Scientist, Fast Code AI - Bangalore

Sep 2025 – Present

- Engineered an interruption-enabled full-duplex voice chat server, orchestrating speech models with tool-calling pipelines and managing **vLLM** audio inference context for real-time conversational interactions
- Researched and fine-tuned full-duplex speech architectures (**Moshi**, **Personaplex**) for domain-specific use cases, benchmarking turn-taking, barge-in robustness, and conversational coherence
- Exploring training of a conversational TTS model targeting natural prosody and contextual expressiveness, with data pipeline built on LLM-adapted transcripts and **OpenAI/Gemini** TTS synthesis
- Integrated **Qwen3-TTS** via **vLLM-Omni** into the speech inference stack, adding asynchronous audio chunking and input text streaming to achieve sub-150ms end-to-end latency
- Drove productionization of the voice AI system through thorough evaluation across latency, interruption handling, contextual understanding, and noise robustness

Associate AI Researcher, Fast Code AI - Bangalore

Oct 2024 – Aug 2025

- **Odo: Depth-Guided Diffusion for Identity-Preserving Body Reshaping (Oral WACV, 2026 [arXiv](#))**
 - Constructed a large-scale dataset of identity-consistent body-shape variations through a multi-stage pipeline involving diffusion-driven generation, precise human segmentation, and background normalization
 - Trained **Odo**, the first end-to-end diffusion-based human reshaping model combining **SMPL depth-map** conditioning with input image feature injection for controllable, identity-preserving body shape manipulation
- Designed and implemented a text-to-motion generation pipeline based on the **Flux diffusion** model architecture, incorporating Rotary Positional Embeddings for improved temporal coherence across motion frames
- Fine-tuned **LoRA adapters** for the **Flux-Fill** model on a large-scale human-clothing dataset for virtual try-on

Software Applications Engineer, Texas Instruments - Bangalore

Aug 2023 – Oct 2024

- Member of the DLP team, focused on debugging embedded systems for customer issue resolution
- Took ownership of supporting customer companies in verifying and refining their hardware and software designs, ensuring successful projector model launches after comprehensive testing
- Improved legacy embedded C code by optimizing, adding new features, and fixing software bugs

Technical Skills

Programming Languages: Python, C/C++ , Java, Julia, MATLAB, Embedded C

Frameworks & Libraries: PyTorch, TensorFlow, Hugging Face, Diffusers, Transformers, Scikit-learn, OpenCV

Tools & Software: Git, Docker, Arduino IDE, Android Studio, Unity, AutoCAD, GNU Radio, Flutter, EAGLE

Projects

Few-Shot Plant Disease Classification 2022 – 2023

Guide Prof. Rajbabu Velmurugan | RnD & B.Tech Project

- Developed a few-shot deep learning model using prototypical networks to classify plant diseases from leaf images, achieving 87% accuracy on unseen disease classes
- Researched prompt learning techniques via context optimization for **CLIP**, adopting them on the PlantVillage dataset to achieve 82% few-shot classification accuracy
- Benchmarked deep learning models for onion crop disease detection, improving performance via data augmentation and sampling to achieve 92.4% classification accuracy
- Integrated trained **PyTorch** models into an Android application using **PyTorch Mobile**, with a real-time image capture pipeline for on-device disease classification

IIT Bombay Student Satellite Program 2020

Instrumentation Subsystem | GLEE - Great Lunar Expedition for Everyone System

A 70-member student team dedicated to making IIT Bombay a center of excellence in Space Technology. GLEE is a global collaborative mission that will conduct research on the lunar surface with Chipsats.

- Implemented I2C communication protocol between a gyroscope sensor and an Arduino UNO
- Simulated lunar seismic models in Python using Devito to implement a finite difference method to solve wave propagation equations to visualize the stress patterns in localized areas

Cancer Detection using CNNs Nov 2021

Guide Prof. Suyash Awate | CS 736 (Medical Image Computing) Course Project

- Developed deep learning models for metastatic tissue detection in histopathologic lymph node scans
- Implemented VGG and custom CNN architectures using Keras and TensorFlow for robust feature extraction and classification, achieving 94% accuracy and 0.97 AUC-ROC on the validation set
- Built image denoising, segmentation, and statistical shape analysis pipelines, enhancing interpretability

RL Agent for Atari Breakout May 2021

Guide Prof. Abir De | CS 419 (Introduction to ML) Course Project

- Trained an agent in the OpenAI Gym environment to play the Cartpole and Atari Breakout game
- Implemented the Deep Q-Learning and Double Deep Q-Learning algorithms in Reinforcement Learning

Image Regularization using PDEs Oct 2021

Guide Prof. Ajit Rajwade | CS 663 (Digital Image Processing) Course Project

- Researched a vector-valued image regularization framework based on partial differential equations (PDEs)
- Implemented image inpainting, reconstruction, denoising, magnification, and flow visualization in MATLAB using the PDE-based regularization technique

3D Object Reconstruction Dec 2021

Guide Prof. Rajbabu Velmurugan

- Developed an image pre-processing pipeline using OpenCV to remove background from object images
- Executed the forward path of the Pix2Vox model to reconstruct 3D objects from a single image

Scholastic Achievements

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| • Secured All India Rank 206 in JEE Advanced out of 180 thousand candidates | 2019 |
| • Achieved All India Rank 463 in JEE Mains out of 1 million students | 2019 |
| • Recipient of the Kishore Vaigyanik Prosthahan Yojana (KVPY) Fellowship from IISc | 2018 |